

Manan Bhatia

Resumé

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Date of birth: 17th January, 1998
Citizenship: Indian

Education

- 09/2026–present **Postdoctoral Researcher**, *University of Geneva*, Host: Hugo Duminil-Copin.
05/2026–08/2026 **Postdoctoral Researcher**, *HKU Mathematics*, Host: Wei Qian.
09/2021–05/2026 **PhD Student**, *MIT Mathematics*, Advisor: Scott Sheffield.
2020 - 2021 **Visiting student**, *ICTS Bangalore*, Supervisor: Riddhipratim Basu.
2016 - 2020 **BSc(Research): Mathematics**, *Indian Institute of Science*, Bangalore.

Research interests

I am interested in probability theory. Currently, I am working on problems related to the random geometries arising from models in the LQG and KPZ universality classes.

Research articles

Published articles

Manan Bhatia. The $d_\gamma/2$ -variation of distance profiles in γ -Liouville quantum gravity. *Commun. Math. Phys.*, 406(3):58, 2025. Id/No 61.

Manan Bhatia. The metric removability of interfaces in the directed landscape. *Annals of Applied Probability*, 36(2):1272–1308, 2026.

Riddhipratim Basu, Manan Bhatia, and Shirshendu Ganguly. Environment seen from infinite geodesics in Liouville quantum gravity. *Ann. Probab.*, 52(4):1399–1486, 2024.

Manan Bhatia. Duality in the directed landscape and its applications to fractal geometry. *Int. Math. Res. Not.*, 2024(12):9633–9662, 2024.

Riddhipratim Basu and Manan Bhatia. Small deviation estimates and small ball probabilities for geodesics in last passage percolation. *Isr. J. Math.*, 264(1):37–96, 2024.

Manan Bhatia, Byron Chin, Nitya Mani, and Elchanan Mossel. When will (game) wars end? *The American Mathematical Monthly*, 133(1):35–46, 2026.

Manan Bhatia. Moderate deviation and exit time estimates for stationary last passage percolation. *Journal of Statistical Physics*, 181(4):1410–1432, Sep 2020.

Preprints

Manan Bhatia. A non-trivial dynamics on the directed landscape. *arXiv:2609.35124*, 2026.

Manan Bhatia. Geodesic traces in dynamical Brownian last passage percolation. *arXiv:2609.35103*, 2026.

Manan Bhatia and Konstantinos Kavvadias. Uniformity of extremal behaviour along geodesics in Liouville quantum gravity. *arXiv:2608.22693*, 2026.

Manan Bhatia, Ewain Gwynne, and Brin Harper. Dimension lower bounds in random geometry via Lipschitz functions. *arXiv:2606.10496*, 2026.

Manan Bhatia and Konstantinos Kavvadias. Strong confluence of geodesics in Liouville quantum gravity. *arXiv:2512.09219*, 2025.

Manan Bhatia. Geodesic switches and exceptional times in dynamical brownian last passage percolation. *arXiv:2510.27589*, 2025.

Manan Bhatia. Near-existence of bigeodesics in dynamical exponential last passage percolation. *arXiv:2504.12293*, 2025.

Manan Bhatia, Ewain Gwynne, and Jinwoo Sung. Area measures and branched polymers in supercritical Liouville quantum gravity. *arXiv:2410.12693*, 2024.

Riddhipratim Basu and Manan Bhatia. A Peano curve from mated geodesic trees in the directed landscape. *arXiv:2304.03269*, 2023. Under revision at Probability and Mathematical Physics.

Manan Bhatia. Atypical stars on a directed landscape geodesic. *arXiv:2211.05734*, 2022. Under revision at Transactions of the American Mathematical Society.

Prizes and awards

- 2024 **MathWorks Science Fellowship.**
- 2024 **Honorable mention for Jane Street Graduate Research Fellowship**, *Jane Street*, New York.
- 2021 **Peterson fellowship**, *MIT Mathematics Department*.
- 2018 **S.N. Bhatt memorial summer fellowship**, *ICTS Bangalore*, Supervisor: Riddhipratim Basu.
- 2016 **Silver medal at the International Physics Olympiad 2016**, *Zurich, Switzerland*.
- 2016 **Honorable mention at the Asian Physics Olympiad 2016**, *Hong Kong*.
- 2015 **Indian National Mathematics Olympiad awardee**, Invited to attend IMO team selection camp.
- 2014 **KVPY fellowship**, National rank 30.
- 2013 **Gold medal at the International Junior Science Olympiad 2013**, *Pune, India*.

Invited Talks

- 07/26 **Hong Kong Probability Workshop 2026**, *The University of Hong Kong*, Hong Kong.
- 03/26 **AMS Spring Eastern Meeting**, *Boston, USA*.
- 10/25 **Stanford Probability Seminar**, *Palo Alto, USA*.
- 09/25 **Columbia Probability Seminar**, *New York, USA*.
- 07/25 **Stochastic Processes and Applications**, *Wrocław, Poland*, Invited session.
- 06/25 **ICTS Probability Seminar**, *Bangalore, India*.
- 11/24 **MIT Probability seminar**, *Cambridge, USA*.
- 06/24 **IISc Analysis and Probability Research Group seminar**, *Bangalore, India*.
- 05/24 **KIAS Analysis, PDE and Probability seminar**, *Seoul, South Korea*.
- 03/24 **KPZ meets KPZ workshop**, *Toronto, Canada*.
- 02/24 **Toronto Probability Seminar**, *Toronto, Canada*.
- 02/24 **UChicago Probability and Statistical Physics Seminar**, *Chicago, USA*.
- 12/23 **Columbia Probability Seminar**, *New York, USA*.
- 10/23 **University of Utah Stochastics Seminar**, *Salt Lake City, USA*.
- 05/23 **Hong Kong Probability Seminar**, *Hong Kong*.
- 03/23 **Infosys Chandrasekharan Random Geometry Colloquium**, *Mumbai, India*, Online.
- 06/22 **PIMS Summer School in Probability**, *Vancouver, Canada*.
- 05/22 **Paris-Saclay Probability seminar**, *Orsay, France*.
- 09/21 **Random networks and interacting particle systems**, Online.
- 07/20 **Bangalore probability seminar**, *Bangalore, India*, Online.